

**HORN, Sweden**WMO: **025570**Lat: **57.8861N**Lon: **15.8623E**Elev: **299**StdP: **14.54**Time Zone: **1.00 (EUC)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -5.5       | 2.8 | -8.9                          | 3.4 | -5.4 | 0.0 | 5.6 | 3.1  | 23.6                  | 41.3 | 20.8 | 41.2 | 0.2                   | 10   | 0.592 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 22.3                   | 80.8            | 65.5 | 77.6 | 64.5 | 74.3 | 62.7 | 68.4                | 76.2 | 66.3 | 74.1 | 64.5 | 71.8 | 6.9                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 65.7                            | 95.8 | 71.4 | 63.6 | 89.0 | 69.6 | 61.5 | 82.6 | 67.6 | 32.8          | 76.3 | 31.1 | 74.3 | 29.7 | 71.8 | 80.1              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |             |             | Extreme Annual Temperature |             |                    |            | n-Year Return Period Values of Extreme Temperature |             |              |             |              |             |              |             |  |  |
|-------------------|-------------|-------------|----------------------------|-------------|--------------------|------------|----------------------------------------------------|-------------|--------------|-------------|--------------|-------------|--------------|-------------|--|--|
|                   |             |             | Mean                       |             | Standard Deviation |            | n=5 years                                          |             | n=10 years   |             | n=20 years   |             | n=50 years   |             |  |  |
| 1%                | 2.5%        | 5%          | Min                        | Max         | Min                | Max        | Min                                                | Max         | Min          | Max         | Min          | Max         | Min          | Max         |  |  |
| (a)               | (b)         | (c)         | (e)                        | (f)         | (g)                | (h)        | (i)                                                | (j)         | (k)          | (l)         | (m)          | (n)         | (o)          | (p)         |  |  |
| (4) <b>18.3</b>   | <b>16.0</b> | <b>14.1</b> | <b>-10.4</b>               | <b>84.9</b> | <b>10.2</b>        | <b>3.0</b> | <b>-17.7</b>                                       | <b>87.1</b> | <b>-23.7</b> | <b>88.8</b> | <b>-29.5</b> | <b>90.4</b> | <b>-36.9</b> | <b>92.6</b> |  |  |
| (5)               |             |             | <b>-11.2</b>               | <b>71.3</b> | <b>9.9</b>         | <b>3.3</b> | <b>-18.3</b>                                       | <b>73.7</b> | <b>-24.1</b> | <b>75.6</b> | <b>-29.6</b> | <b>77.4</b> | <b>-36.8</b> | <b>79.8</b> |  |  |

**Monthly Climatic Design Conditions**

|      |                                                                                  | Annual      | Jan    | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                                  | (d)         | (e)    | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 43.3   | 26.7  | 27.6  | 33.1  | 41.1  | 49.8  | 56.8  | 61.1  | 59.4  | 52.0  | 43.6  | 36.9  | 30.8  | (6)  |
| (7)  |                                                                                  | DBStd       | 13.98  | 10.23 | 9.53  | 8.75  | 6.20  | 5.88  | 5.11  | 4.68  | 4.71  | 5.53  | 7.04  | 7.03  | 9.92  | (7)  |
| (8)  |                                                                                  | HDD50       | 3479   | 723   | 626   | 525   | 273   | 76    | 6     | 0     | 2     | 41    | 217   | 394   | 596   | (8)  |
| (9)  |                                                                                  | HDD65       | 7945   | 1188  | 1046  | 990   | 718   | 471   | 251   | 140   | 184   | 389   | 664   | 843   | 1061  | (9)  |
| (10) |                                                                                  | CDD50       | 1043   | 0     | 0     | 0     | 5     | 71    | 210   | 344   | 292   | 102   | 18    | 1     | 0     | (10) |
| (11) |                                                                                  | CDD65       | 33     | 0     | 0     | 0     | 0     | 0     | 5     | 19    | 9     | 0     | 0     | 0     | 0     | (11) |
| (12) |                                                                                  | CDH74       | 761    | 0     | 0     | 0     | 0     | 28    | 168   | 405   | 151   | 9     | 0     | 0     | 0     | (12) |
| (13) |                                                                                  | CDH80       | 123    | 0     | 0     | 0     | 0     | 1     | 23    | 77    | 22    | 0     | 0     | 0     | 0     | (13) |
| (14) | Wind                                                                             | WSAvg       | 5.9    | 6.3   | 6.2   | 6.3   | 5.8   | 5.8   | 5.8   | 5.4   | 5.5   | 5.8   | 5.9   | 6.2   | 6.3   | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 24.3   | 1.4   | 1.3   | 1.1   | 1.3   | 2.0   | 2.7   | 3.5   | 2.9   | 1.8   | 2.2   | 2.2   | 2.0   | (15) |
| (16) |                                                                                  | PrecMax     | 29.3   | 2.5   | 2.4   | 1.7   | 3.2   | 3.5   | 5.2   | 6.5   | 5.9   | 4.6   | 4.7   | 4.1   | 3.0   | (16) |
| (17) |                                                                                  | PrecMin     | 20.2   | 0.2   | 0.3   | 0.1   | 0.1   | 0.6   | 0.9   | 1.1   | 0.4   | 0.3   | 0.6   | 0.6   | 0.7   | (17) |
| (18) |                                                                                  | PrecStd     | 2.6    | 0.7   | 0.5   | 0.4   | 0.9   | 0.8   | 1.1   | 1.3   | 1.4   | 1.1   | 1.1   | 1.0   | 0.7   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 47.1  | 49.3  | 59.4  | 69.3  | 77.3  | 82.8  | 85.7  | 82.3  | 74.7  | 64.5  | 54.0  | 51.9  | (19) |
| (20) |                                                                                  |             | MCWB   | 44.5  | 44.5  | 47.7  | 54.4  | 60.1  | 64.6  | 66.7  | 66.6  | 64.7  | 56.9  | 51.3  | 48.9  | (20) |
| (21) |                                                                                  | 2%          | DB     | 43.5  | 45.1  | 54.4  | 64.4  | 72.9  | 78.8  | 82.1  | 78.0  | 70.3  | 59.7  | 51.3  | 48.4  | (21) |
| (22) |                                                                                  |             | MCWB   | 41.1  | 41.6  | 45.5  | 51.6  | 57.9  | 63.3  | 66.1  | 67.2  | 61.5  | 55.5  | 49.3  | 45.6  | (22) |
| (23) |                                                                                  | 5%          | DB     | 41.3  | 42.4  | 49.2  | 59.6  | 69.1  | 75.0  | 78.7  | 74.3  | 66.5  | 57.0  | 49.1  | 45.2  | (23) |
| (24) |                                                                                  |             | MCWB   | 39.1  | 39.4  | 43.0  | 49.2  | 56.4  | 61.7  | 65.2  | 64.6  | 58.9  | 53.4  | 47.3  | 43.0  | (24) |
| (25) |                                                                                  | 10%         | DB     | 39.0  | 39.9  | 45.5  | 55.5  | 65.0  | 71.3  | 75.1  | 71.3  | 63.1  | 54.6  | 46.8  | 42.4  | (25) |
| (26) |                                                                                  |             | MCWB   | 37.3  | 37.5  | 41.0  | 46.6  | 54.5  | 59.1  | 64.0  | 62.5  | 57.1  | 51.6  | 45.3  | 40.5  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 45.1  | 45.5  | 49.6  | 56.5  | 63.2  | 68.3  | 70.4  | 73.0  | 66.3  | 58.4  | 52.1  | 49.2  | (27) |
| (28) |                                                                                  |             | MCD B  | 46.9  | 48.0  | 56.5  | 68.3  | 72.5  | 77.1  | 79.5  | 79.0  | 72.9  | 61.3  | 53.1  | 51.5  | (28) |
| (29) |                                                                                  | 2%          | WB     | 41.6  | 42.1  | 46.3  | 52.8  | 60.4  | 64.9  | 68.4  | 68.6  | 62.6  | 56.0  | 49.6  | 45.9  | (29) |
| (30) |                                                                                  |             | MCD B  | 43.2  | 44.7  | 53.4  | 61.9  | 68.8  | 74.5  | 77.1  | 74.9  | 68.3  | 59.2  | 51.1  | 48.2  | (30) |
| (31) |                                                                                  | 5%          | WB     | 39.3  | 39.7  | 43.6  | 49.8  | 57.9  | 62.6  | 66.8  | 65.9  | 60.0  | 53.9  | 47.5  | 43.1  | (31) |
| (32) |                                                                                  |             | MCD B  | 40.9  | 42.1  | 48.4  | 57.6  | 66.1  | 73.5  | 75.5  | 71.8  | 64.9  | 56.7  | 48.9  | 44.9  | (32) |
| (33) |                                                                                  | 10%         | WB     | 37.2  | 37.5  | 41.3  | 47.4  | 55.4  | 60.3  | 64.9  | 64.0  | 58.1  | 51.9  | 45.4  | 40.7  | (33) |
| (34) |                                                                                  |             | MCD B  | 38.9  | 39.8  | 45.1  | 54.3  | 63.5  | 69.1  | 73.1  | 69.6  | 62.2  | 54.2  | 46.6  | 42.2  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | MDBR        |        | 11.6  | 13.6  | 17.9  | 22.8  | 24.2  | 23.4  | 22.3  | 21.0  | 20.0  | 15.2  | 10.0  | 10.4  | (35) |
| (36) |                                                                                  | 5% DB       | MCD BR | 10.7  | 14.4  | 24.5  | 32.2  | 33.4  | 32.5  | 30.7  | 27.4  | 25.7  | 18.4  | 11.9  | 10.5  | (36) |
| (37) |                                                                                  |             | MCWBR  | 9.7   | 11.5  | 16.8  | 20.5  | 19.6  | 18.3  | 16.4  | 17.4  | 17.3  | 14.0  | 10.5  | 9.7   | (37) |
| (38) |                                                                                  | 5% WB       | MCD BR | 10.4  | 12.6  | 22.3  | 27.9  | 27.6  | 28.8  | 25.6  | 23.9  | 23.6  | 16.5  | 11.2  | 10.5  | (38) |
| (39) |                                                                                  |             | MCWBR  | 9.7   | 10.4  | 15.9  | 18.5  | 17.7  | 17.4  | 15.0  | 16.3  | 16.7  | 13.3  | 10.1  | 9.8   | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.286 | 0.320 | 0.338 | 0.376 | 0.373 | 0.362 | 0.389 | 0.381 | 0.349 | 0.340 | 0.313 | 0.274 | (40) |
| (41) |                                                                                  | taud        |        | 2.269 | 2.316 | 2.361 | 2.324 | 2.397 | 2.451 | 2.396 | 2.413 | 2.487 | 2.430 | 2.317 | 2.210 | (41) |
| (42) |                                                                                  | Ebn at Noon |        | 180   | 227   | 258   | 264   | 273   | 278   | 267   | 260   | 253   | 219   | 167   | 145   | (42) |
| (43) |                                                                                  | Edh at Noon |        | 15    | 23    | 28    | 34    | 34    | 33    | 34    | 31    | 25    | 20    | 15    | 12    | (43) |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 113   | 305   | 781   | 1277  | 1667  | 1824  | 1649  | 1324  | 897   | 407   | 148   | 80    | (44) |
| (45) |                                                                                  | RadStd      |        | 18    | 57    | 82    | 152   | 145   | 89    | 185   | 129   | 116   | 68    | 28    | 9     | (45) |